AivaCare - Agentic AI Continuous Care Platform

Solution Architecture Document

Client	LiverWell Clinic
Domain	Fatty liver disease (MASLD / MASH)
Version	1.0 - roadshow prototype
Date	1 November 2026
Status	Prototype on synthetic data. Not for clinical use.

1. Problem Statement

LiverWell Clinic treats patients with fatty liver disease using an **appointment-based model**: a hepatology visit every three months and periodic blood tests. Many patients also track their own health on wearables.

Example patient: John Miller, 48, diagnosed with MASLD

- He sees his hepatologist every 3 months and has periodic lab tests.
- His Apple Watch tracks activity, heart rate, sleep and weight every day.
- His medical records, lab results and wearable data sit in **separate systems that don't share data**.

For the patient	For the care team
Can't make sense of health trends between visits	No visibility into progress between appointments
No personalized guidance when questions come up	Concerning trends are noticed late, so interventions are delayed
Hard to stick to diet, activity and medication plans	Coordinating follow-ups by hand takes a lot of effort

Goal: move LiverWell to a **continuous care model**. Specialized AI agents monitor patients proactively, give personalized support, and coordinate timely clinical interventions. **Physicians stay in control, and clinical safety is guaranteed.**

Success measures

- Patient engagement and treatment adherence

- Time from a concerning finding to clinician acknowledgment
- Percentage of escalations resolved within the SLA
- Clinical trends across the patient population: weight, ALT/AST, FIB-4, HbA1c
- Hours saved on routine care coordination

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2. Solution Overview

AivaCare is a secure, multi-agent AI platform with three entry points: a **patient mobile app**, a **patient web portal**, and a **clinician care dashboard**.

Design principles

#	Principle	What it means
1	**Rules decide, AI explains, physicians approve**	Deterministic rules detect clinical risk. AI interprets the finding and communicates it. Clinicians make every clinical decision.
2	**Agents only where judgment is needed**	Two AI agents handle conversation and reasoning. Everything else is a deterministic service or a single-purpose AI tool.
3	**Facts from records, knowledge from approved sources**	Numbers come straight from the clinical record. Guidance comes only from clinician-approved content, and answers cite their sources.
4	**Governed access**	Agents reach data and actions only through controlled tool servers (MCP), scoped to the signed-in user, with every call audited.
5	**Standards-based**	HL7 FHIR R4 is the system of record, so the platform can integrate with any modern EMR.

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3. Capabilities

Capability	What it delivers	Delivered by
Unified Patient 360	One record over time that combines EMR, lab, history and wearable data	Ingestion service → HAPI FHIR repository
Medical Document Intelligence	Patients upload reports; AI extracts, validates and summarizes them	Claude document extraction + validation + patient confirmation
AI Patient Assistant	Personalized answers grounded in the patient's own record and approved knowledge, with sources cited	**Patient Care Agent** + RAG

Capability	What it delivers	Delivered by
Continuous Monitoring	Checks incoming data and trends against thresholds set by clinicians	Deterministic rules engine (versioned rule catalog)
Clinical Escalation	Prioritized alerts with SLAs, tracked until resolved	LangGraph escalation workflow + FHIR Task
Clinician Insights	AI alert reviews, Patient 360, Copilot Q&A, draft care-plan changes	**Clinician Copilot Agent** + alert-review tool
Patient Engagement	Reminders, follow-up messages, progress summaries	Notification service + follow-up writer tool
Governance & Safety	Human approval, guardrails, audit trail, red-flag bypass	Guardrails, audit log, approval interrupts

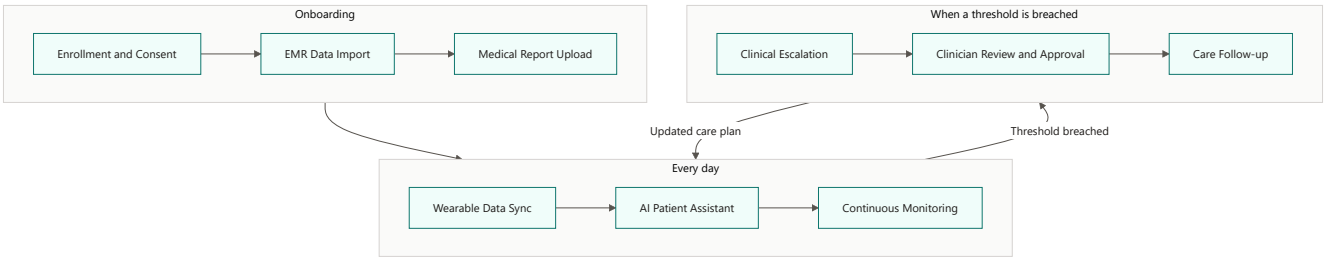
Why two agents and not more

- **One agent per trust boundary.** The Patient Care Agent handles untrusted patient input and can only touch that patient's data. The Clinician Copilot serves authenticated clinicians and reaches only their assigned panel.
- **Safety-critical logic stays deterministic.** Detection, escalation, SLAs and the red-flag path are rule-based, auditable and unit-tested.
- **Lower risk and cost.** Fewer AI decision points mean less evaluation and validation effort, lower latency and a lower cost per interaction.

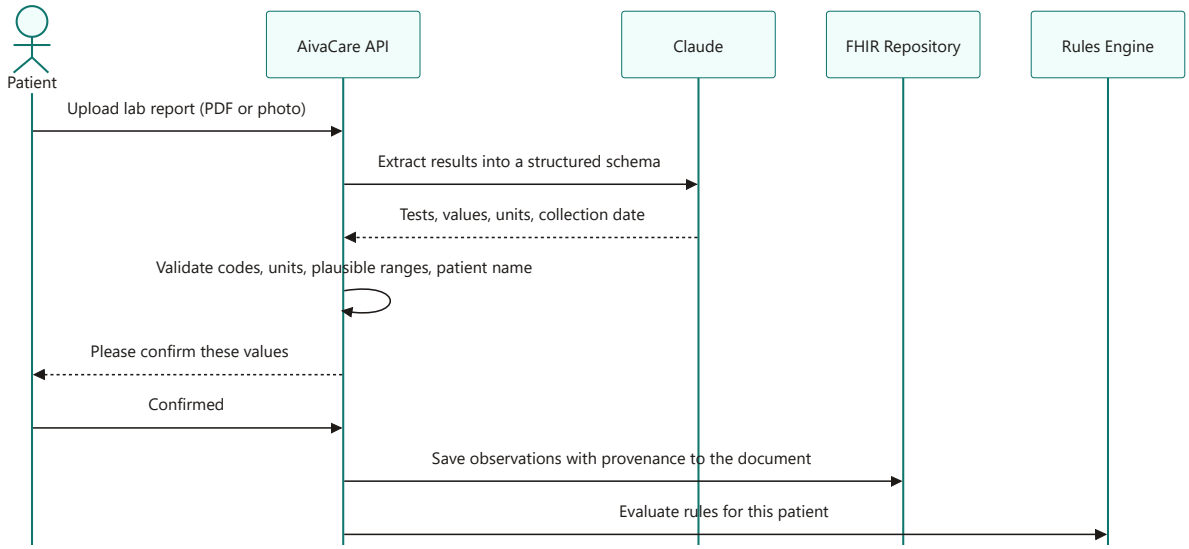
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4. Process Flows

4.1 End-to-end patient journey

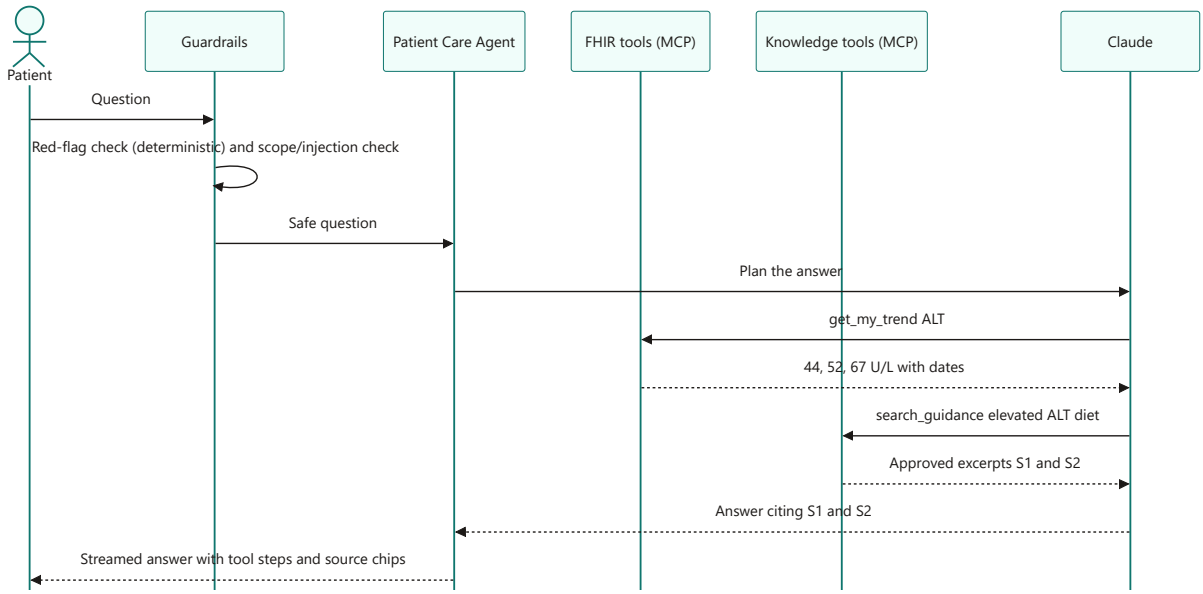


4.2 Medical report upload



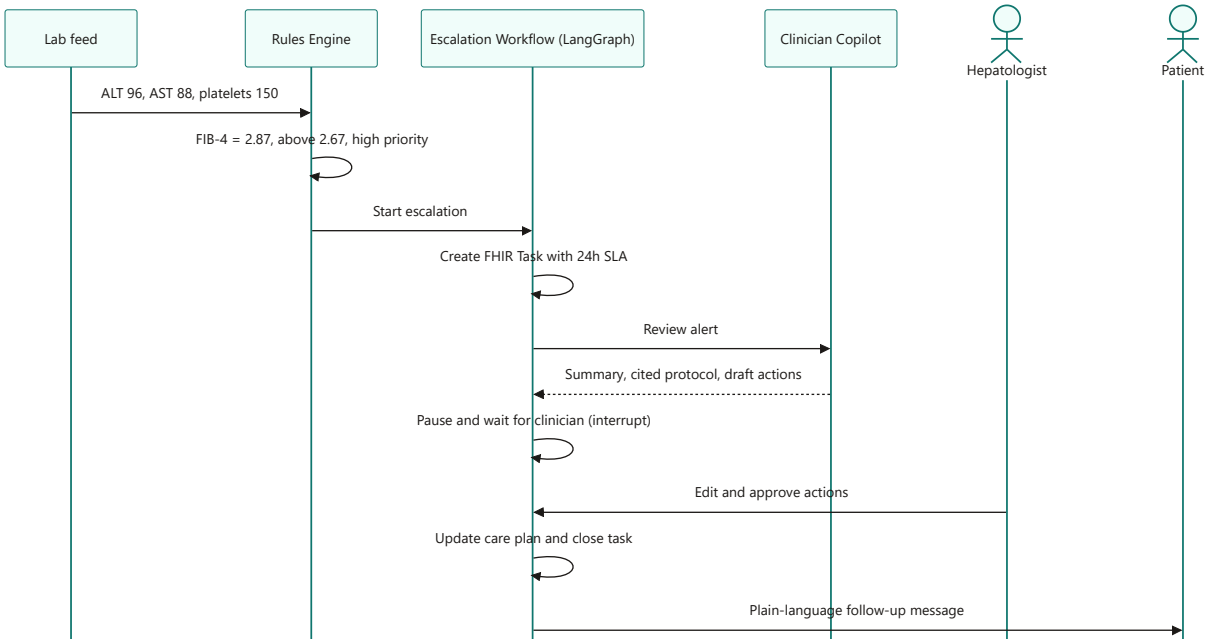
4.3 Grounded patient question and answer

Example question: "*Why is my ALT high and what should I eat?*"

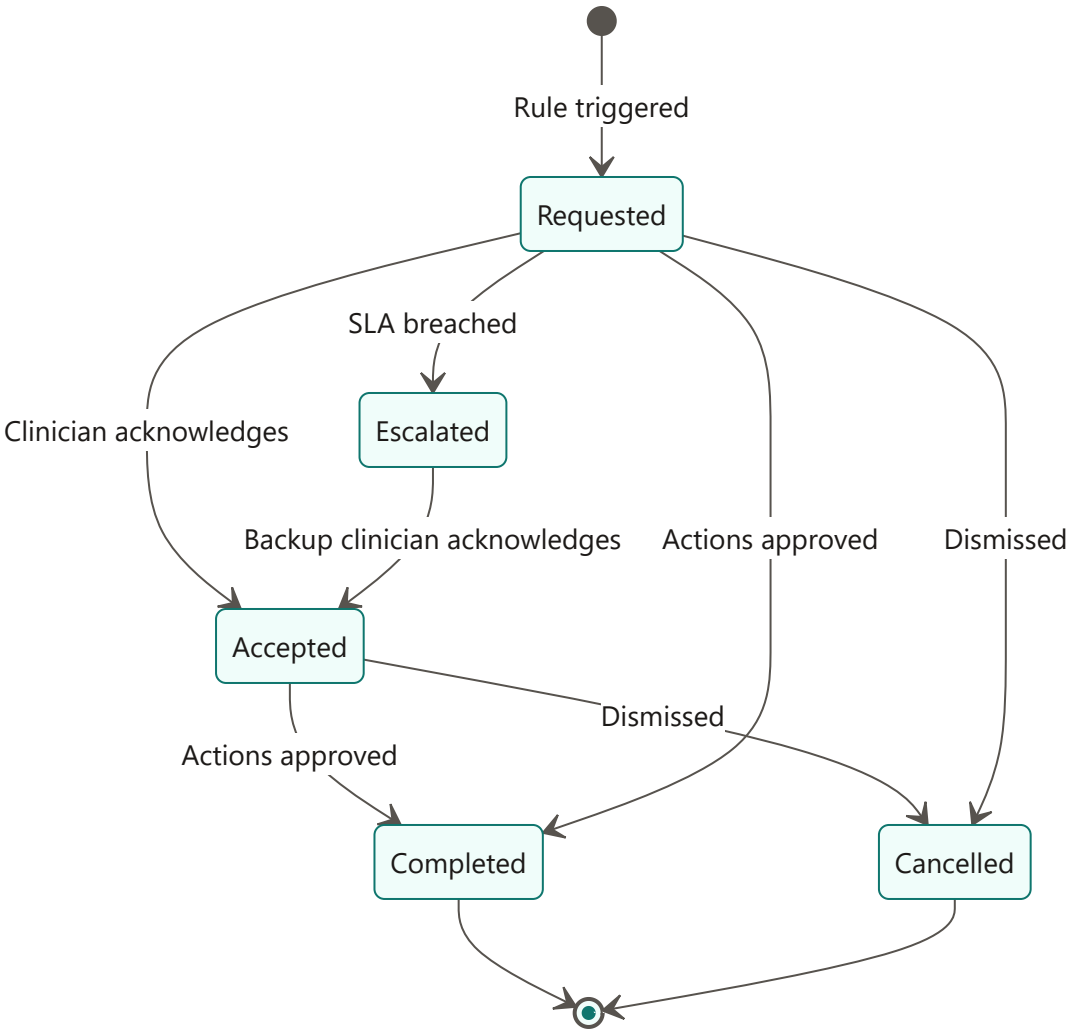


If a **red-flag symptom** is detected (yellow eyes or skin, vomiting blood, black stools, confusion, abdominal swelling), the AI is bypassed. The patient immediately gets urgent-care instructions and a high-priority escalation is created.

4.4 Monitoring → escalation → clinician approval → follow-up



4.5 Escalation lifecycle

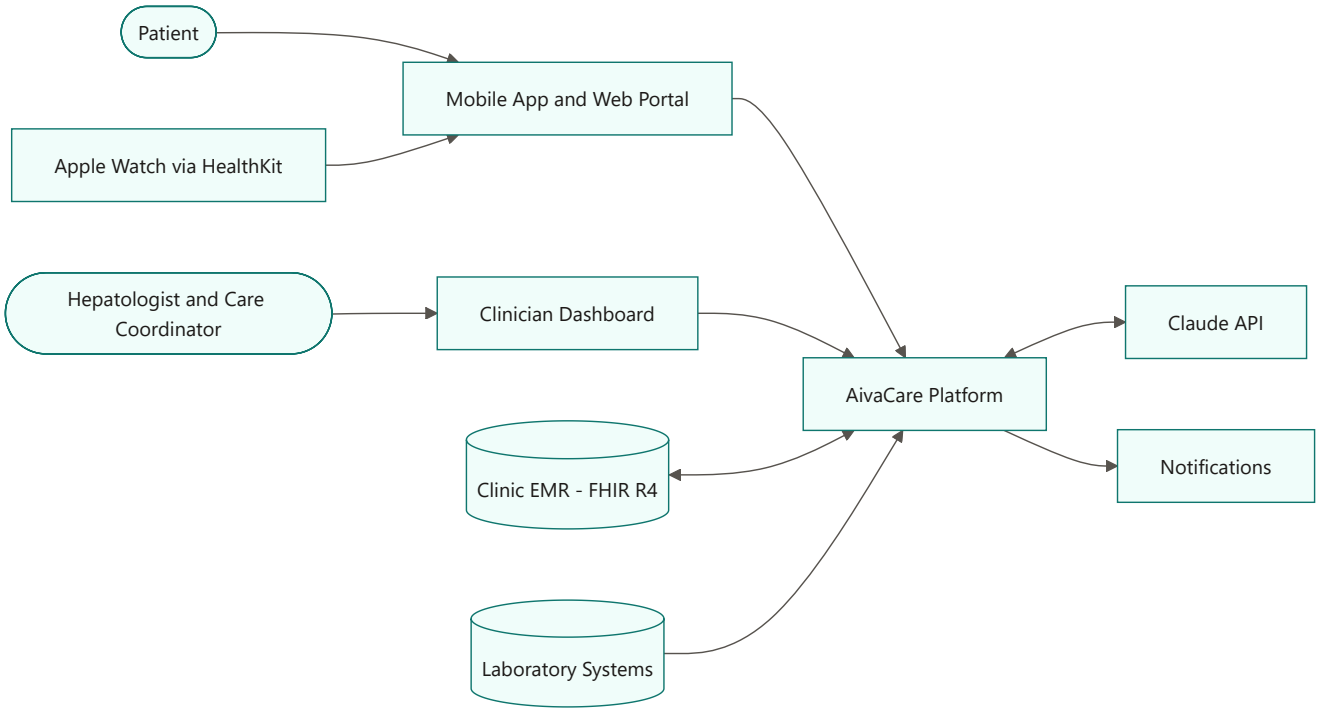


Each escalation is stored as a FHIR **Task**, with a priority, an owner, an SLA deadline, the rule evidence, the AI review, the clinician's decision and an audit trail. The workflow state is checkpointed in Postgres, so a pending approval survives restarts. SLA re-escalation is planned for the pilot phase.

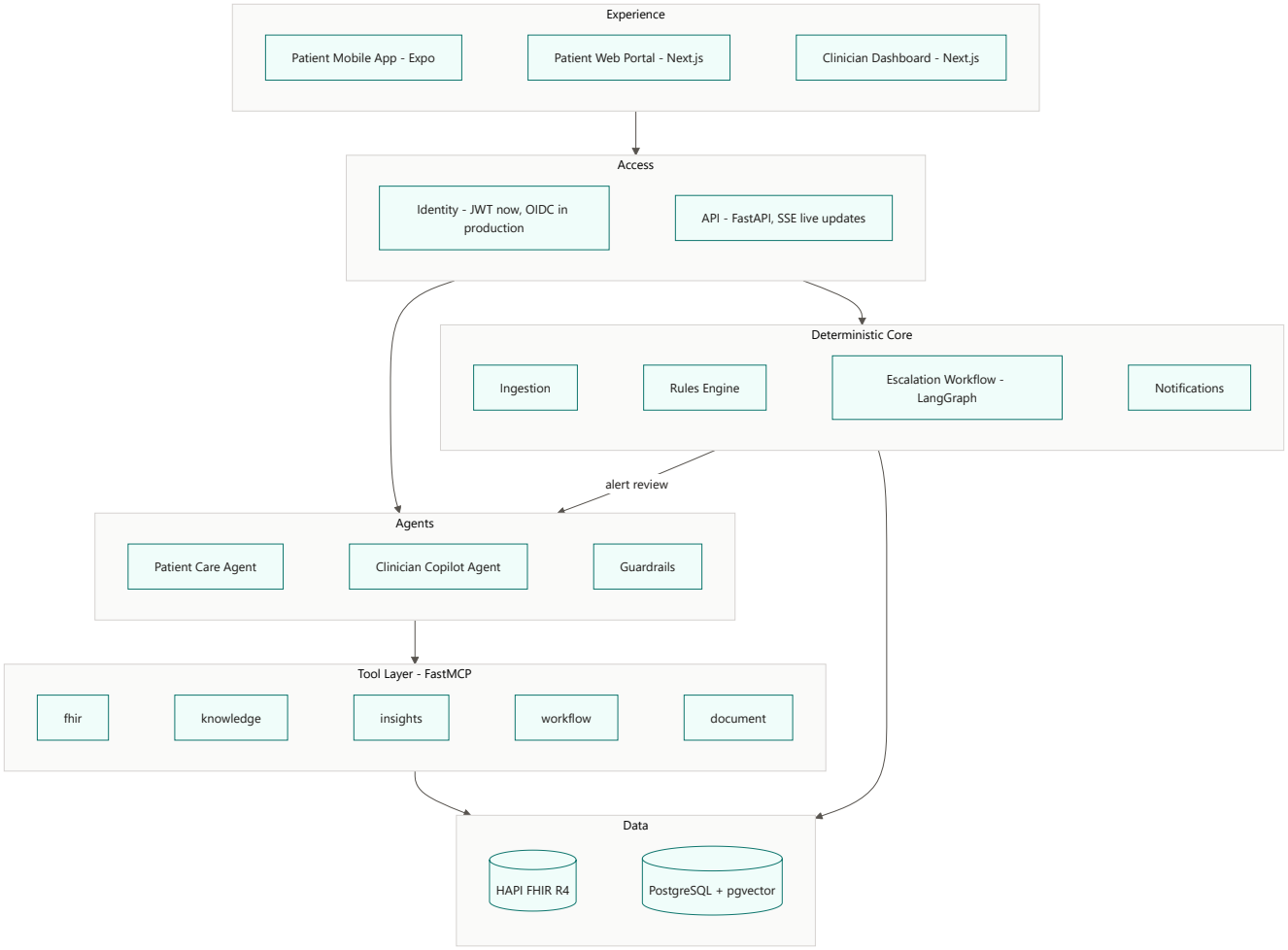
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5. System Landscape

5.1 Context diagram



5.2 Layered architecture



5.3 Technology stack

Layer	Technology
Patient mobile app	React Native / Expo, streaming chat over `expo/fetch`
Web portal and dashboard	Next.js, Tailwind CSS, Recharts, Vercel AI SDK UI (streaming chat)
Backend	Python 3.12, FastAPI, Server-Sent Events
LLM	Claude API via the official Anthropic SDK: Claude Opus 5.5 (agents, reviews, extraction) and Claude Haiku 5.5 (guardrail classification, follow-up messages), with server-side refusal fallback
Agent orchestration	Streaming Claude tool loop; **LangGraph** for the escalation workflow (Postgres checkpointer, `interrupt()` for human approval)
Tool integration	**FastMCP**: five servers composed per request, called through an in-process MCP client

Layer	Technology
RAG	PostgreSQL + **pgvector**, local embeddings (BAAI bge-small via fastembed), hybrid vector + full-text search
Interoperability	**HAPI FHIR R4** with LOINC, SNOMED CT and RxNorm coding
Rules	Python rules engine over a versioned YAML rule catalog
Infrastructure	Docker Compose, Caddy (automatic HTTPS), GitHub

5.4 Agents

Users	Patients (untrusted input)	Hepatologists and care coordinators
Purpose	Q&A, education, coaching, symptom logging, appointment requests	Alert review, Patient 360 questions, protocol lookup, draft care-plan changes
Data scope	The signed-in patient only	The assigned patient panel
Can write	Symptom logs, appointment requests, confirmed report data	Care-plan *proposals* only; applying them needs clinician approval
Never	Diagnoses, changes medication, handles emergencies on its own	Makes clinical changes without physician sign-off

Single-purpose AI tools (one structured call each, evaluated in isolation):

- **Alert review:** summary, cited protocol and draft actions.
- **Document extraction:** lab report → validated schema.
- **Follow-up writer:** plain-language message to the patient.
- **Guardrail classifier:** in scope, out of scope, or prompt injection.

5.5 Tool layer (MCP)

Server	Patient tools	Clinician tools
fhir	`get_my_profile`, `get_my_trend`, `get_my_latest_results`, `get_my_care_plan`	`list_patient_panel`, `get_patient_summary`, `get_trend`
knowledge	`search_guidance` (patient content)	`search_guidance` (clinician + patient content)
insights	`get_my_progress`	`get_risk_flags` (rules engine)

Server	Patient tools	Clinician tools
workflow	`log_symptom`, `request_appointment`	`list_open_alerts`, `propose_careplan_change`
document	`list_my_documents`	`list_patient_documents`

Security controls in the tool layer

- **Identity comes from the login token, never from the AI.** Patient tools have no patient-ID parameter, so a prompt cannot request another patient's data.
- **Panel checks.** Clinician tools verify that the patient is in the clinician's panel.
- **Proposals, not changes.** Clinical writes become drafts that a clinician must approve.
- **Full audit.** Every consequential action is written to the audit log.

5.6 Knowledge and RAG

Information	Source	Retrieval
Labs, vitals, wearables, medications, care plan	FHIR repository	**Exact query through tools** (never similarity search)
Patient education, clinic protocols	Clinician-approved markdown with version and approver metadata	**RAG** filtered by audience, with a similarity threshold

When no approved content matches, the assistant says so and offers to contact the care team instead of guessing.

5.7 Clinical rule catalog (demo values; clinical sign-off required)

Rule	Threshold	Priority / SLA
FIB-4 high	> 2.67	High / 24h
FIB-4 indeterminate	1.3 - 2.67	Medium / 72h
ALT above 3× upper limit of normal	> 120 U/L	High / 24h
ALT doubled from the patient's baseline	≥ 2.0×	Medium / 72h
Weight gain	> 3% in 30 days	Low / 7d
Resting heart rate rise	> 8 bpm vs. baseline	Low / 7d
Activity drop	> 50% vs. baseline	Low / 7d
Red-flag symptoms	Keyword patterns	High / immediate patient instructions

5.8 Security, privacy and compliance

Area	Prototype	Production
Identity	Demo personas, signed JWT, role checks; only physicians can approve or dismiss clinical actions (coordinators triage and acknowledge)	OIDC SSO (e.g. Keycloak or Identity Platform), MFA for clinicians, SMART on FHIR launch from the EMR
Data protection	Synthetic data only, HTTPS	Encryption at rest with customer-managed keys, PHI redaction in logs, minimum-necessary data sent to the LLM
AI safety	Red-flag bypass, scope and injection classifier, citations, draft-only clinical writes	Plus evaluation suites in CI, red-team tests, sampled clinician review
Secrets and LLM credentials	Claude API key in a root-only env file on the VM; workspace spend cap; access-code gate with lockout after repeated failures; cost guards (chat rate and length limits, upload rate and page limits, daily AI call budget with non-AI fallbacks)	Workload Identity Federation (short-lived tokens issued to the service's cloud identity, no static API keys); Secret Manager for remaining secrets
Audit	Audit log table	FHIR AuditEvent and Provenance, immutable log storage
Regulatory	Not applicable (demo)	HIPAA safeguards and BAAs (hosting and LLM provider); non-device CDS positioning under FDA guidance; ONC HTI-1 transparency

6. Deployment

6.1 Roadshow prototype

Item	Choice
Hosting	One GCP Compute Engine VM running Docker Compose in a dedicated `aivacare` project
Services	HAPI FHIR, PostgreSQL + pgvector, API, web, Caddy (HTTPS)
Mobile	Android preview APK (EAS); Expo Go and the responsive web portal as fallbacks
Cost control	Start the VM before demos and stop it afterwards (only disk is billed when stopped); rate limits and a spend cap on the Claude API
Fallback	The same Compose stack runs on the presenter's laptop

6.2 Production target (GCP)

Component	Service
Web and API	Cloud Run (autoscaling)
FHIR store	Cloud Healthcare API (FHIR R4) or managed HAPI on GKE
Database	Cloud SQL for PostgreSQL with pgvector
Events	Pub/Sub, FHIR notifications
Secrets	Secret Manager
Observability	Cloud Logging and Monitoring, Langfuse or OpenTelemetry for agent traces
LLM	Claude via the Anthropic API or Vertex AI, under a BAA

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7. Implementation Steps

7.1 Completed: roadshow prototype

Step	Delivered
1. Foundation	Docker Compose stack, HAPI FHIR, Postgres/pgvector, synthetic data (John Miller's 12-month story + 8 panel patients), wearable simulator
2. Patient 360 and Assistant	MCP tool servers, RAG over approved content, streaming Patient Care Agent with citations, guardrails
3. Safety core and HITL	Rules engine, event-driven escalations, LangGraph workflow with approval interrupt, Clinician Copilot, live alert queue
4. Document AI and engagement	Report upload → extraction → validation → confirmation → FHIR; follow-up messages and notifications
5. Experience	Patient web portal, clinician dashboard, presenter controls, Expo mobile app

7.2 Next: pilot (about 3 months)

Phase	Focus	Exit criteria
Discovery (2 weeks)	Clinical sign-off of the rule catalog and content; EMR and lab integration design; security review	Signed-off scope and rules

Phase	Focus	Exit criteria
Integration (4 weeks)	EMR FHIR API, lab HL7 v2 feed, HealthKit sync in a native iOS build, OIDC SSO	Real data flowing in a test environment
Hardening (4 weeks)	Evaluation suites in CI, SLA re-escalation, observability, PHI controls, penetration test	Eval and security gates pass
Pilot (4+ weeks)	25-50 consenting patients; clinician feedback; tuning thresholds	Engagement and response-time targets met

7.3 Production

- HIPAA-eligible hosting and BAAs.
- Clinical safety case and hazard log.
- Disaster recovery.
- Scale to the full patient population.
- Extend to related conditions such as diabetes and obesity.

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8. Risks, Assumptions and Open Questions

Risk	Mitigation
Unsafe or inaccurate AI advice	Grounding in approved content only, citations, guardrails, red-flag bypass, eval gates
Alert fatigue	Tiered priorities, clinician-tunable thresholds, one task per patient event, AI summaries for context
Document extraction errors	Validation rules, patient confirmation, provenance
Cross-patient data exposure	Identity bound into tools, panel checks, audit
Regulatory classification	Non-device CDS positioning; a physician makes every clinical decision

Assumptions

- The prototype uses synthetic data and simulated wearables.
- Clinical content and rule thresholds are demo values that need clinician review.

Open questions for LiverWell

1. Which EMR do you use, and does it offer FHIR R4 API access?
2. How are lab results delivered today (HL7 v2, FHIR or a portal)?
3. Who triages alerts, during which hours, and with what response times?
4. Which therapies and protocols are in scope?
5. How many patients will be in the pilot and in full rollout?
6. Do you have a preferred hosting environment?